Abiodun Victor Alagbada

Software Engineer | PhD Candidate in Applied Mathematics



Budapester Straße 2, 99427 Weimar, Germany
+49 176 2654 7570 | abiodunalagbada@gmail.com
abiodun.victor.alagbada@uni-weimar.de
linkedin.com/in/abiodunalagbada | github.com/Alagbada123
alagbada123.github.io | ORCID 0000-0003-4801-8907

Profile

Doctoral researcher in applied mathematics at Bauhaus-Universität Weimar and certified back-end software engineer. My doctoral work builds data-driven and physics-informed models for structural health monitoring, with a peer-reviewed systematic review on Koopman operator methods published in 2026. I build Python and JavaScript services, have led a Scrum team on an industry pipeline-orchestration project, and hold UK Qualified Teacher Status after more than ten years of mathematics teaching in Nigerian schools and online.

Professional Experience

Full-Stack and Back-End Developer 2022 – Present
Freelance

- Build web applications and REST APIs in Python (Django, Flask, FastAPI) and JavaScript (Node.js, React).
- Package services with Docker and automate testing and deployment through GitHub Actions.
- Produce the YouTube series *Full-Stack Journey with Abiodun Alagbada*, teaching web development to beginners.

Online Mathematics Tutor 2020 – Present
Preply and Upwork

- Design individual study plans in Mathematics and Advanced Mathematics for learners in several countries.
- Teach with Jupyter, marimo and LaTeX materials; published an interactive advanced-mathematics notebook on molab.marimo.io.

Mathematics and Computer Science Teacher 2018 – 2023
Omega High School, Jos, Nigeria

- Taught Mathematics, Advanced Mathematics and Computer Science, and supervised the ICT laboratory.
- Brought programming and data visualisation into lessons; led assessment moderation and curriculum planning.

Teacher of Mathematics, Advanced Mathematics and Physics 2014 – 2017
Corpus Christi Comprehensive College, Jos, Nigeria

- Planned and delivered senior secondary lessons in Mathematics, Advanced Mathematics and Physics.
- Prepared students for West African Senior School Certificate examinations.

Leadership

Scrum Master, GATE Industry Project with MKP GmbH (Marx Krontal Partner) WS 2025/26
Bauhaus-Universität Weimar

- Led an interdisciplinary, intercultural team under the Scrum framework on *Prototypical Pipeline Orchestration for Engineers*, a task set by MKP GmbH.
- Steered delivery of a real-time pipeline platform built on Vue 3, FastAPI, WebSockets, Redpanda, QuixStreams, the Docker SDK and GitHub Actions.
- Completed 70 hours of project work and the Agile Project Management workshop, and presented an academic project poster.

Thesis mentoring: structured written feedback on an MSc thesis in progress, *Bauhaus-Universität Weimar* 2026

National Youth Service Corps, one-year national service, *Plateau State, Nigeria* Nov 2014 – Nov 2015

Professional memberships: Associate Member, Institute of Strategic Entrepreneurs, Nigeria (since 2015); Mathematical Association of Nigeria; Black in Mathematics; American Mathematical Society.

Education

PhD in Applied Mathematics (in progress) May 2024 – expected 2028
Bauhaus-Universität Weimar, Institute of Structural Mechanics. Supervisor: Prof. Tom Lahmer

Research area: Koopman operator methods and hybrid physics–data modelling for vibration-based structural health monitoring.

MSc Mathematics 2016 – 2018 (awarded Nov 2019)
University of Jos, Nigeria

Thesis: *A Class of Block Second Derivative Method for Solving Stiff Ordinary Differential Equations.* Supervisor: Prof. Geoffrey Kumleng.

Postgraduate Diploma in Education (PGDE) Admitted 2017 (awarded Sep 2020)
University of Ilorin, in affiliation with the National Teachers' Institute, Kaduna, Nigeria

BSc Mathematics, Second Class Honours (Upper Division) 2010 – 2014
University of Jos, Nigeria

Thesis: *The Convergence of Some Complex Power Series.* Supervisor: Dr. Silas Longwap.

Further Education and Certification

Software Engineering Programme (12 months), Back-End Specialisation, *ALX Africa*
May 2024 – Jul 2025

Finite Element Method and Computational Structural Dynamics (12 weeks, 4 credits), *NPTEL, India* Jul – Oct 2026

Certificate in Agile Project Management (GATE Programme), *Bauhaus-Universität Weimar*
Jan 2026

Qualified Teacher Status (TRN 2361519), *Department for Education, England* Mar 2025

Registered Certified Teacher (Reg. No. PL/R/07163), *Teachers Registration Council of Nigeria*
Since Dec 2022

Web Development, *Opecstat Academy* Mar 2024

Web Design Consultancy Course, *Nova University* 2023

Back-End Web Development (Node.js), Internship Cohort 6, *Side Hustle* Nov 2022

Soft Skills Training, *Jobberman* Sep 2022

Research and Publications

Doctoral research. I develop interpretable, data-driven models of structural dynamics for condition assessment and damage diagnosis, combining Koopman operator theory, system identification and physics-informed learning in reproducible Python workflows.

Journal articles and book chapters

- A. V. Alagbada and T. Lahmer. Koopman operator methods in structural health monitoring: A systematic review towards hybrid physics–data frameworks. *Applied Sciences*, 16(7):3392, 2026. doi:10.3390/app16073392
- A. Alagbada, T. C. Iroko and S. Tchoneteck. Machine learning: Theoretical foundations, practical applications, ethical implications, and future prospects. Exploring challenges and applications in emerging technologies. In M. Khosrow-Pour (ed.), *Encyclopedia of Modern Artificial Intelligence.* IGI Global, 2027. doi:10.4018/407565

Preprints

- A. Alagbada, S. E. Fadugba, G. Kumleng, A. Bankole and M. Kekana. Singularity-induced stability barriers in block second-derivative collocation methods for stiff initial value problems. SSRN preprint. doi:10.2139/ssrn.7289249

- T. Iroko, A. Alagbada and S. Tchoneteck. Neural regime-switching model with Markov-informed attention for short-horizon equity return forecasting. Preprints.org, 2026. doi:10.20944/preprints202603.1462.v1

Technical Skills

Scale: basic working expert

Programming

Python
JavaScript
C/C++
SQL
MATLAB

Web and Back-End

Django / Flask
FastAPI
Node.js
React / Vue 3
REST and WebSockets

Scientific Computing and ML

NumPy / Pandas
TensorFlow
scikit-learn
Jupyter / marimo

Tools and Delivery

Git / GitHub
Docker
GitHub Actions
LaTeX / TikZ
Scrum

Languages

English (native) • Yoruba (native) • German (A2–B1)

Honors and Awards

DAAD Doctoral Scholarship, *German Academic Exchange Service* 2025

PTDF Scholarship, *Petroleum Technology Development Fund, Nigeria* 2023/2024 cohort

Certificate of Excellence (Bronze Medal), NAMCUS 2013, *National Mathematical Centre, Abuja* Feb 2013

Fourth Place, Singles, University Table Tennis Tournament, *Bauhaus-Universität Weimar* Dec 2025



Abiodun Victor Alagbada
Weimar, November 6, 2026